

- 4 (a) (i) *either* lines directed away from sphere
or lines go from positive to negative
or line shows direction of force on positive charge M1
so positively charged A1 [2]
- (ii) *either* all lines (appear to) radiate from centre
or all lines are normal to surface of sphere B1 [1]
- (b) tangent to curve B1
in correct position and direction B1 [2]
- (c) (i) $V = (0.76 \times 10^{-9}) / (4\pi \times 8.85 \times 10^{-12} \times 0.024)$ C1
= 285 V A1 [2]
- (ii) negative charge is induced on (inside of) box M1
formula applies to isolated (point) charge
OR less work done moving test charge from infinity A1
so potential is lower A1 [3]
- (d) *either* gravitational field is always attractive
or field lines must be directed towards both box and sphere B1 [1]

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5 (a) e.g. concrete charges, store energy, smoothing circuit, etc.

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